

Examiner reports

Q1.

Part (a) was well attempted and many students scored full marks. The question left students with the choice of how to approach the proof and inefficient but correct methods were often seen. Students who scored one mark often failed to complete the proof. Students should be in the habit of finishing any proof or “show that” question with a concluding statement which completes their argument.

While many students scored full marks on part (b) the most efficient method was rarely seen. Students who solved $p(x) = 0$ on their calculators and made use of the factor theorem to write down the correct factorisation could achieve full marks in one line.

Part (c) discriminated well between students of varying abilities. Many students could start the question by manipulating the given equation to form a cubic equation. Some students who went on to see the connection with the earlier part of the question made errors with their use of inequalities or failed to complete their argument with enough rigor.

Some students seemed to focus on the phrase “no real solutions” and were determined to try and use the discriminant of an incorrect quadratic.

Q4.

In part (a), apart from a small number of students who treated the expression as a product of terms instead of a sum and hence gained no marks, this was usually answered correctly. Where there were errors, it was usually with the first term as the students wrote e^{3x} or $3e^x$ instead of $3e^{3x}$.

In part (b)(i), most students applied the quotient rule correctly. However, poor notation such as $1 + \cos^2 x$ instead of $(1 + \cos x)^2$ or $\sin X^2$ instead of $\sin^2 x$ and sign errors caused marks to be lost. There were also examples of false cancellation leading to the correct result. In part (b)(ii), here the students tended to do either very well or very badly, scoring 2 or 0. Those students attempting to use $y = \ln(\sin x) - \ln(1 + \cos x)$ rarely made progress.

Q5.

In Part (a), most students gained the method mark by using the identity $\sec^2 x = 1 + \tan^2 x$ and many went on to earn the accuracy marks. A significant number of students lost the final accuracy mark through poor notation such as writing $\tan x^2$ instead of $\tan^2 x$. It was

surprising to see some students working at this level to write $\frac{\sec^2 x}{\sec^2 x - 1}$ as $\frac{\sec^2 x}{\sec^2 x} - \frac{\sec^2 x}{1}$.

Part (b) proved difficult for most students. Many students assumed that they could factorise the quadratic equation and they were satisfied with factors that were clearly incorrect, such as $(\operatorname{cosec} x - 3)(\operatorname{cosec} x + 1)$. Some students misquoted the quadratic formula; some others muddled factors and roots. Most students showed a correct conversion from $\operatorname{cosec} x$ to $\sin x$ after obtaining their roots but those who attempted to convert their equation at the start rarely proceeded to a correct quadratic equation in $\sin x$. Some students discarded the negative root of their quadratic equation.

In part (c), the students who answered part (b) correctly usually gained both marks for this part, although some lost the accuracy mark for not writing their answers to the nearest degree. Where both marks were not earned, the majority of the students gained the method mark for equating $2\theta = 60^\circ$ to one of their solutions in part (b).

Q6.

The majority of students did well on part (a) of this question and well over half of the cohort scored 6 marks or more. Most students were less successful on part (b), although many started by writing down several relevant looking identities.

Part (a)(i). The vast majority of students answered correctly, with relatively few finding $\alpha = 56.3^\circ$ rather than the expected $\alpha = 33.7^\circ$.

Part(a)(ii). Although most students scored the 3 marks some failed to state the minimum value where an explicit statement was required. Similarly, some students just wrote down the value of x at which the minimum occurred without justifying it. Less common errors

were to use $\cos^{-1}\left(-\frac{1}{R}\right)$ or to assume the minimum was at $x = 90^\circ$.

Part (b)(i). Many students wrote the given expression in terms of $\sin x$ and $\cos x$, but then went on to treat the identity as if it were an equation, often equating to zero. Those who used a common denominator or took out a factor of $\frac{1}{\sin x}$ usually completed successfully.

Part (b)(ii). Although most students recognised that they had to use the result from part (i), many failed to consider both $\cot x = 0$ and $\cos 2x = 0$ and so gained no credit. Of those who did consider both these equations many students thought $\cot x = 0$ had no solutions, or they changed it to $\tan x = 0$. Most students did solve $\cos 2x = 0$ correctly, although not always giving both solutions within the given range.

Q7.

Many students appeared to have covered this part of the specification very superficially. Many students earned no marks for this question because they did not appreciate that the functions were only valid over a restricted domain. Students should be aware that when co-ordinates are requested these should be stated separately as an ordered pair; in this instance credit was given for correct values marked on the axes, but this was a generous concession.

(a) Some students lost the sketch mark because their gradients were wrong or their curves did not reach the x -axis. A small number of students had the y values as $\frac{\pi}{2}$ or 2π . Some students drew the cosine graph or the graph of $\frac{1}{\cos x}$ while a significant proportion did not attempt this part of the question.

(b) All the above issues surfaced again in this part. Some of the students who earned the marks for part (a) carried out wrong transformations.

Q8.

This question was poorly answered.

In part (a), although many students achieved full marks, some found the combining of two fractions beyond them. For many students, an initial step was replacing $1 + \cos \theta$ with $\sin \theta$ or attempting to multiply through by $1 + \cos \theta$, thus obtaining expressions such as

$$1 + \frac{1 + \cos \theta}{1 - \cos \theta} = 32$$

Despite the majority of students finding part (b) of this question difficult, full marks were occasionally seen. Students who were unsuccessful in part (a) were able to attempt this part using the result given. Most students earned the first method mark, but for many this was all they achieved since they only worked with $\sin \theta = +\frac{1}{4}$ and then only worked to 2dp which resulted in just 2 solutions. For the final two B marks, the solution $x = 0.17$ was often missing, and 2.00 was often written as 1.99.

Q9.

Part (a) proved to be a good source of marks, although a significant number of students lost a mark through poor notation, such as $\cos y^2$ or no $\frac{dx}{dy}$

Although many earned full marks in part (b), many other proofs were inadequate. Students were expected to use the identity $\tan^2 y + 1 = \sec^2 y$ and to show the expansion of $(x - 1)^2$ clearly. Too many fudged their working, wrote $\tan y + 1 = \sec y$ or only proved the known trig identity.

The result in part (c) was easily derived from the previous two parts, but there were many blanks here and many confused statements about inverse functions which were often

taken as $\frac{1}{\tan y}$. Students who used the formula book appropriately generally earned the mark.

There were many fully correct solutions to part (d)(i). However it was distressing at this stage to see some horrendous algebra, with terms inverted in ways that were invalid and worthless.

Only the more able students were able to make progress on the last two parts of the question. Many students differentiated their quadratic instead of the expression for $\frac{dy}{dx}$. Some students earned the method mark in the final part, but few completed the whole answer, as their final explanation often had an element missing.

Q10.

In part (a)(i) it was clear that dx/dy confused many, they simply wrote down $dy/dx =$. Many solutions were correct, but some candidates missed the multiplier and some were clearly trying to use the product rule or split $3y$ and $+1$, which was an indication of poor understanding of work on trigonometric functions.

Part (a)(ii) was not very well answered by many candidates. Those who had managed to get the

method mark in part (a)(i) often gained the method mark for obtaining $\frac{dy}{dx} = 3\sec^2(0)$.

However,

many incorrectly rearranged the function to $\frac{dy}{dx} = \frac{1}{3\cos^2(3y+1)}$ not $\frac{\cos^2(3y+1)}{3}$, so substituting into an incorrect function

In part (b) a large number of candidates had no idea of the shape of the inverse tangent graph. A few incorporated a turning point and many failed to put the correct values on the y -axis. For the correct shapes that were seen, the y -axis was often not labelled, so only part marks were awarded. The main error seen was a reflection of $y = \tan x$ in the y -axis and not in $y = x$.

Q11.

In part (a) most candidates were able to obtain the first angle of 1.77 radians and hence obtain 2 marks. Many candidates also obtained the correct second solution, although 4.91 was a common error.

Part (b) was the worst answered question part on the paper, with very few candidates having any idea of how to go about combining the two fractions. Very few completely correct solutions

were seen. Those who managed to reduce the LHS to $\frac{-2\operatorname{cosec}^2 x}{-\cot^2 x}$ could often not complete the question. It was sad to see some very poor algebraic techniques in evidence here. There were many cases where candidates did not attempt the question.

In part (c) far too many candidates failed to recognise both the positive and the negative roots for $\sec x$ and in consequence they did not score any marks, since they only found the solutions for $\cos x = 0.2$.

Q12.

In part (a)(i), although the majority of candidates scored both marks here some rounded to the nearest degree, some truncated, some rounded 14.477 to 14.8, some only considered +4, and

some erroneously added -14.5 or 14.5 to 90 . A handful took cosec to be $\frac{1}{\tan}$ or $\frac{1}{\cos}$.

In (a)(ii) there were some excellent full solutions here but also some disturbing misunderstanding of notation. Many candidates found it disconcerting to deal with the functions of $(2x + 30)$, for example $2\operatorname{cosec}^2 x - 1 + (2x + 30) = 2 - 7\operatorname{cosec}(2x + 30)$ was seen. They would have been well advised to use a substitution such as Y for $(2x + 30)$. A few missed the brackets around $(1 - \operatorname{cosec}^2 Y)$ thus getting a wrong equation and no further marks; some had this correct but failed to deal with the constants correctly and again could not progress. Sadly the incorrect substitution of $\cot Y + 1$ for $\operatorname{cosec} Y$ was also seen on several occasions. Once the correct factors were obtained a few stopped there (particularly those who unfortunately substituted x for $2x + 30$). Quite a few handled the -4 solution well but not the $\frac{1}{2}$ as they went on to assume $\sin(2x + 30)$ was $\frac{1}{2}$ and not 2 .

In part (b) most candidates recognised that the required transformations were a stretch and a translation. The majority started with the stretch and then, wrongly took the translation to be $[-30,0]$ instead of $[-15,0]$. A few missed the correct term “stretch” and many failed to use the right terminology of “scale factor” $\frac{1}{2}$. A few had the direction of the stretch in the y -axis or the SF as 2 and a few had their translation in the y -direction.

Q13.

In part (a)(i), even though the sketch was often poorly drawn, many candidates obtained full marks, and, where they did not, it was often because sketches went beyond the correct end points. Some correct graphs had the wrong end points marked, reversed coordinates being the most common error. There were however many candidates who had no idea what the graph looked like.

Not all candidates attempted part (a)(ii). Where lines were drawn, they were often not accompanied by sufficient explanation to award the accuracy mark. There were also many instances of lines with a positive gradient intersecting in the third quadrant.

In part (b), where candidates clearly defined which function they were using many achieved full marks. There are still candidates who lose marks by saying “change of sign so the root lies between these two values” without stipulating $0.5 < x < 1$.

Part (c)(i) was usually well answered, with answers given to the required degree of accuracy.

Part (c)(ii) was very well answered with most candidates obtaining both marks.

Q14.

- (a) The majority of candidates drew a modulus graph and therefore earned the method mark. Many correct responses to the number of sections were seen although some candidates appeared to try to sketch $y = |\sec^{\frac{1}{2}}x|$. Most candidates who earned the first two marks went on to earn the final 1 mark although poor sketching did lose the 1 mark for variable heights in some cases.
- (b) This was very well answered, with most candidates obtaining both marks. Where marks were lost it was usually through only offering the answer of 60° . A few candidates thought $\sec = 1/\tan$ or $1/\sin$ and hence obtained no marks.

Q15.

Part (a) was well answered by the majority of candidates. $\sec x$ being changed to $\sin x$ was a common error. The first angle was usually found correctly but the second angle was often incorrect, with 318 being a common error. There were very few cases of angles in radians being found.

In part (b), most candidates used the correct identity and were successful in answering this part of the question. The main error was the use of $\tan^2 x = \sec^2 x + 1$, but $5 \sec x = 5(\tan x + 1)$ was also seen. For those who successfully reached the correct quadratic, the factorisation, or use of the quadratic formula, was usually well done, though there were some sign errors. For the final four answers, many totally correct solutions were seen. A common error at this stage was 284 degrees, from $180 + 104$, and not the correct answer

of 256. There were a number of candidates who failed to correctly round their answers to the nearest degree, with $104.47 = 104.5 = 105$ seen.

Q16.

Candidates were generally successful in part (a), sometimes aided by the allowances made for $(-1, 180)$.

Part (b) was well answered by many candidates. Common errors were the use of 'shift' and 'transformation' to describe the word 'translation', and the stretch was often given a scale factor of $\frac{1}{2}$ in the x -axis.

In part (c), the shape of the graph was usually correct, but many candidates lost this mark by going into other quadrants. The mark for the labelling of the axis was not so readily earned, as these values were often missing or incorrect.

In part (d)(i), many candidates gave the correct response. The major error was to state that $2y = \cos^{-1}(x - 1)$ for the first step resulting in no marks being awarded.

In part (d)(ii), most candidates at least obtained the method mark. Where candidates correctly differentiated, they went on to achieve full marks.

Q17.

Part (a) was well answered by the majority of candidates. $\tan x = 0.5$ and $x = 0.46, 3.60$ was a common error. The second angle was often incorrect. Few cases of angles in degrees were seen.

In part (b), most candidates used the correct identity and were successful in answering this part of the question. The main error was $\operatorname{cosec}^2 x = \cot^2 x - 1$ followed by fudging of the rest of the solution.

In part (c), most candidates attempted to factorise the quadratic expression, although some used the quadratic formula. Those who factorised were usually correct, although solutions of $\frac{1}{2}$ and -2 were not uncommon. Those who used the formula often made the error of $\cot^2 x = 2$ or $\cot^2 x = -0.5$. Candidates with 3.60 as a solution in (a) were able to recover here but often failed to obtain both marks; 5.17 was a common error.

Q18.

In general this question was done well by candidates of all abilities. Part (a) was very well answered by the majority of candidates, although $\tan x = 1/3$ and $\sin x = 1/3$ were also seen. 1.23 was usually seen but the second result was often incorrect. There were very few cases of results given in degrees.

Most candidates used the correct identity part (b) in and were successful in answering this part of the question. The main error was candidates using $\tan^2 x = \sec^2 x + 1$ and then fudging the rest of the solution.

In part (c) most candidates attempted to factorise the result from part (b), although some used the quadratic formula. Those who factorised were usually correct, although solutions of $-1/3$ and 1 were not uncommon. For the final 3 answers many totally correct solutions were seen. Candidates with an incorrect solution in part (a) were able to recover here

from follow through marks, but often candidates failed to obtain both marks since 0 or 6.18 often accompanied 3.14.

Q19.

This question was well answered by the majority of candidates. Common incorrect answers involved the use of 3 rather than $1/3$ or naming the y -axis rather than the x -axis when referring to the stretch. The answer of “squash – scale factor 3 – (in the x -direction)” did occur on more occasions than had been expected. The word “transformation” was often used instead of “translation” and the vector for the translation was often described in words rather than stated as a vector.

Q20.

Part (a) had a mixed response. Quite often answers were expressed as decimals and equally frequently in degrees, with and $A = 180$ and $B = 90$ being common errors. Part (b) was a good source of marks for many candidates. Those candidates who correctly arranged the equation as $\cos^{-1} x - 3x - 1 = 0$ were usually successful in answering this question. Problems arose for candidates who tried to rearrange to other forms such as $x = \cos(3x + 1)$, and then were unable to explain their results. Part (c) was usually very well answered, with most candidates receiving full marks.

Q21.

Part (a) was answered very well by the majority of candidates. Where marks were lost, it was generally for an incorrect or omitted second angle in the interval.

Part (b) was also very well answered, with few incorrect responses.

Q22.

- (a) There were few completely correct attempts, with some candidates giving reversed coordinates, and several “coordinates” given without brackets
- (b) Candidates frequently gained both method marks but lost the accuracy mark since their graphs tended to cross the y axis rather than touch it. Where candidates lost method marks it was usually for a translation in the y direction

or for translating $\begin{pmatrix} -1 \\ 0 \end{pmatrix}$.

Q23.

Part (a) Those candidates who appreciated $\sec x = 1/\cos x$ were usually able to evaluate the first answer of 1.37 radians. When candidates went on to complete this part correctly although 4.51 was a common error from $x + 1.37$. Answers in degrees were not common but they were seen.

Part (b) This part was answered very well by the majority of candidates when the most appropriate trigonometrical identity was used when candidates changed the equation into one involving $\sin x$ and $\cos x$, errors often did occur.

Part (c) Most candidates were able to successfully factorise the quadratic and many went on to complete the question correctly. Marks were lost by 'extra' values within the range being given. Candidates were not penalised for extra answers outside the given range.

Q24.

Candidates should know the trigonometrical identities as outlined in the specification. Part (a) was well answered and there were many correct solutions to part (b), although some scripts did not give fully convincing arguments. Part (c) required answers correct to the nearest degree, but answers to a greater degree of accuracy were not penalised. There were a significant number of candidates who obtained an answer of 109.47 which was rounded to 109.5 which was then rounded to 110.

Q25.

This question was the least well answered question on the paper with very few correct solutions. Although the graphs of inverse trigonometrical functions is stated in the specification there were many incorrect graphs sketched. Many candidates had their calculators in degree mode and in part (b)(ii) found the values of 19 and 21 and then stated, 'change of sign, therefore root'. There were similar mistakes in part (c) with iterations jumping from 0.8 to 44 with candidates apparently not realising there was something amiss.



EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.